MovieLens Recommendation System Project

HarvardX PH125.9x - Data Science: Capstone

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1. Introduction

Recommendation systems represent a cornerstone of machine learning applications in the data science era, powering companies like Amazon and Netflix to analyze massive user rating datasets and predict preferences for targeted suggestions. These systems leverage collaborative filtering and matrix factorization to identify patterns in user-item interactions. By processing explicit ratings and implicit behaviors, they enable personalized recommendations that enhance user engagement across digital platforms.

In 2006, Netflix offered a challenge: improve the recommendation algorithm by 10% for a 1 million dollars prize. On September 21, 2009, the grand prize was awarded to the BellKor's Pragmatic Chaos team which bested Netflix's own algorithm for predicting ratings by 10.06% [1]. The Netflix Challenge reflects the high significance of algorithm for products recommendation systems.

The original challenge dataset is not publicly open; instead, in this project, I will use the 10M version of MovieLens dataset, given by GroupLens Research [2], to develop a recommendation system. The dataset will be splited into training set and testing set and the objective is to predict ratings on the test set with a root mean square error (RMSE) of less than 0.86490.

This report presents exploratory data analysis using standard visualization techniques, followed by the methods of developing, training and testing the recommendation algorithm. Results from each model iteration are then provided, concluding with an assessment of the final model's performance, limitations, and opportunities for future enhancements.

In this project, the code was run, and the report was compiled using R Markdown in RStudio Version 2025.09.2+418.

2. Exploratory Data Analysis

2.1 Preparation

The MovieLens 10M dataset (movielens) was downloaded and is partitioned into training and testing set. The training set (edx) contains 90% of the movielens data, while the testing set (final_holdout_test) 10%.

2.2 Overview on the dataset edx

The edx is a data frame that consists of 9,000,055 rows and 6 columns, the column names are:

```
## [1] "userId"    "movieId"   "rating"    "timestamp" "title"     "genres"
```

and it contains ratings from 69,878 unique users for 10,677 unique movies. The average rating across all users and movies is 3.5124.

The distribution of ratings is as follow:

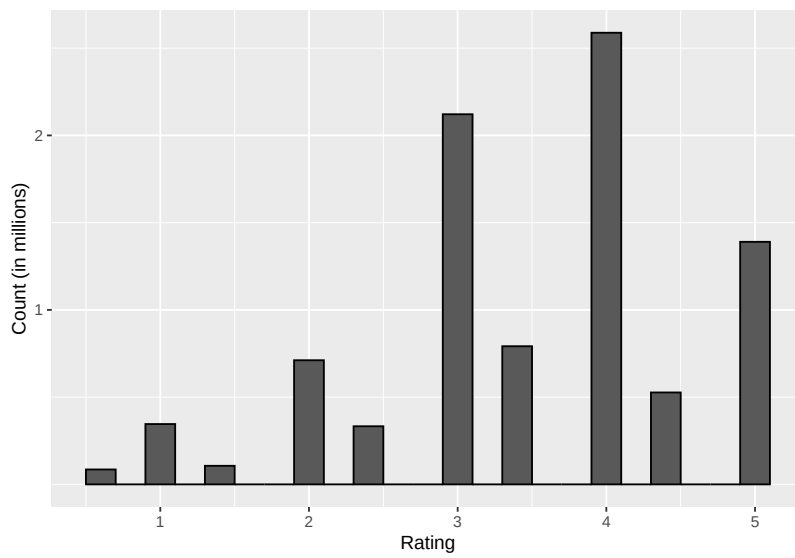


Figure 1: Distribution of Ratings

2.3 Average rating by Movie

Unsurprisingly, some movies are just generally rated higher than others. Figure 2 below shows there is a peak at about average rating = 3.5. Further analysis as in figure 3 reveals an obvious variation in the number of ratings received by a single movie. Pulp Fiction (1994) received the most ratings at 31,362 ratings; while 126 movies garnered just a single rating each. This disparity would stem from popularity trends where heavily rated movies attract more user attention, creating a feedback loop. Thus, a clear movie effect influenced ratings, warranting adjustment in algorithm training.

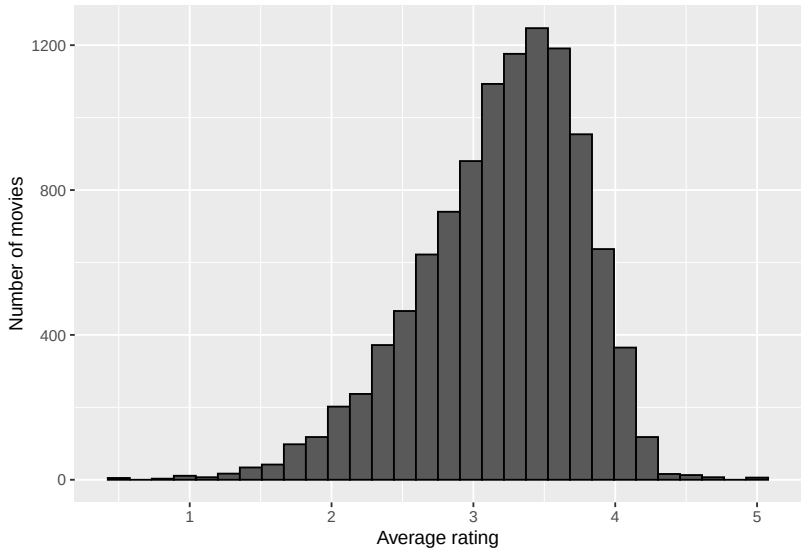


Figure 2: Distribution of Average Ratings by Movie

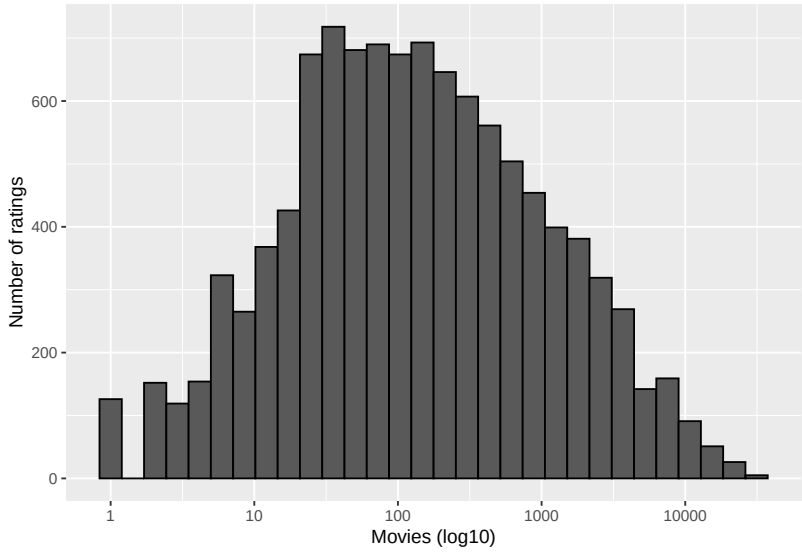


Figure 3: Number of Ratings by Movie

2.4 Average Rating by User

Exploring in terms of user data shows a similar picture as in movie data. A peak at around rating = 3.5 is observed in figure 4. Figure 5 reveals significant variation in user rating activity. The most active user provided 6,615 ratings while 3,470 users contributed fewer than 20 each. This disparity shows frequent raters, likely more engaged, are overrepresented in averages, skewing predictions toward their preferences, whereas rare raters add sporadic, less reliable data. Such user heterogeneity demonstrates a clear user effect, necessitating adjustments in algorithm training.

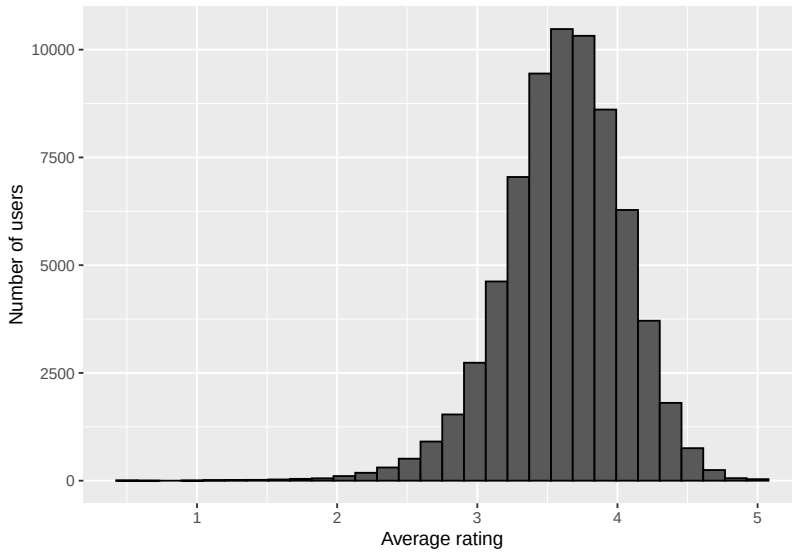


Figure 4: Distribution of Average Ratings by User

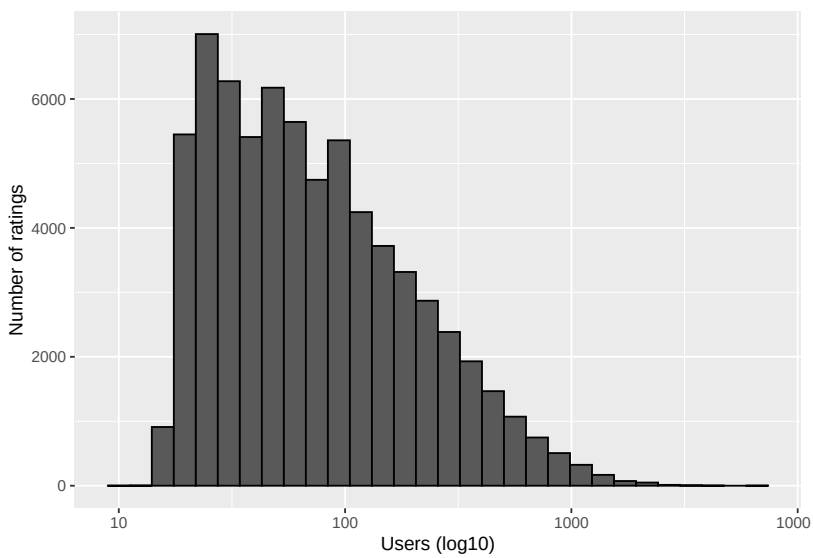


Figure 5: Number Ratings by User

2.5 Average Rating by Movie Genres

The dataset also includes genre information for each movie, with some movies classified under multiple genres. For instance, Titanic (1997) is associated with both Drama and Romance genres. There are 20 primary genre categories in the dataset (see table below), which combine to form a total of 797 unique genre combinations. The average ratings across these basic genre indicate a genre effect, where certain genres tends to receive higher ratings than others. Further analysis of genre combinations with at least 100,000 ratings (see figure 6 below) confirms this variation in rating distributions across combinations. These patterns leads to genre adjustments in algorithm training to account for inherent effects.

Primary Genre	Average Rating	Number of Ratings
Film-Noir	4.012	118541
Documentary	3.783	93066
War	3.781	511147
IMAX	3.768	8181
Mystery	3.677	568332
Drama	3.673	3910127
Crime	3.666	1327715
(no genres listed)	3.643	7
Animation	3.601	467168
Musical	3.563	433080
Western	3.556	189394
Romance	3.554	1712100
Thriller	3.508	2325899
Fantasy	3.502	925637
Adventure	3.494	1908892
Comedy	3.437	3540930
Action	3.421	2560545
Children	3.419	737994
Sci-Fi	3.396	1341183
Horror	3.270	691485

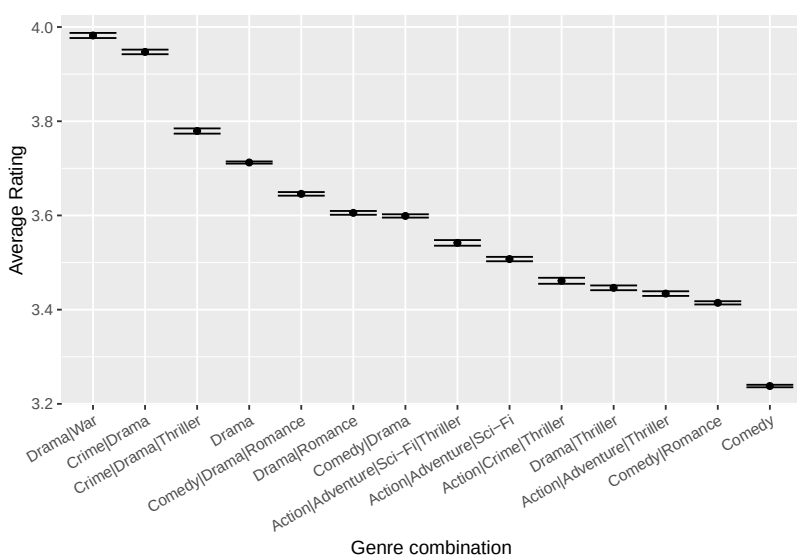


Figure 6: Ratings Distribution by Genre Combinations

3. Methods

3.1 Evaluation of the Algorithm: Residual Mean Square Error as Loss Function

In machine learning, a loss function is used to measure model performance by calculating the deviation of a model's predictions from the correct. Optimizing a model entails adjusting model parameters to minimize the output of some loss function [3]. The Netflix challenge defined the winning entry as the one that minimized the Root Mean Squared Error (RMSE) on a test set [4]. RMSE is calculated by taking the square root of Mean Square Errors (MSE):

$$RMSE = \sqrt{\frac{1}{N}\sum_{u,i}(y_{ui} - \hat{y}_{ui})^2}$$

where y_{ui} denotes the observed rating given by user u to movie i in the test set and $\hat{y}_{ui}$ is the corresponding prediction generated from only the training set, N is the number of ratings in the test set and the sum runs over those ratings. The objective of this project is to develop an algorithm that achieves an RMSE below 0.86490.

3.2 Overall Average Rating Model

The simplest prediction model is using the overall average rating, the mean rating across all users and movies. The model for this is:

$$Y_{u,i} = \mu + \epsilon_{ui}$$

where μ represents the true average rating for all movies and users and ϵ_{ui} represents independent errors with mean zero. This simplest model assumes the same ratings for all movies and users with all differences explained by random variation.

The value of μ that minimizes the RMSE is the least squares estimates, which in this case is simply the average of all ratings in the training set. Therefore, $\hat{\mu}$ = mean(edx$rating) is the simplest formula to train the first algorithm.

Method	RMSE
Overall rating average	1.061202

3.3 Movie Effect Model

In fact, the ratings were not assigned equally across all movies as described in the simplest model. Section 2.3 above identified a significant movie effect influencing ratings, i.e. some movies are generally rated higher than others. The previous model is hence refined to incorporate the movie effect on rating on movie i, b_i:

$$Y_{u,i} = \mu + b_i + \epsilon_{ui}$$

The least squares estimates of the movie effect, $\hat{b}_i$, is the average of $y_{u,i}$ - $\hat{\mu}$ for movie i. So this formula is used to take account of movie effects in training algorithm:

$$\hat{b}_i = mean(\hat{y} - \hat{\mu})$$

Method	RMSE
Overall rating average	1.0612018
Movie Effect	0.9439087

3.4 Movie and User Effect Model

Exploratory data analysis (section 2.4 above) also shows a strong user effect on ratings, i.e. frequent raters are overrepresented in averages, skewing predictions toward their preferences, whereas rare raters add sporadic, less reliable data. The model is further refined by adding user effect on rating from user u, b_u:

$$Y_{u,i} = \mu + b_i + b_u + \epsilon_{ui}$$

Similar to movie effect above, the least squares estimates is obtained by the average of $y_{u,i}$ - $\hat{\mu}$ - b_i. Below formula accounts for the user effects during algorithm training:

$$\hat{b}_u = mean(\hat{y} - \hat{\mu} - \hat{b}_i)$$

Method	RMSE
Overall rating average	1.0612018
Movie Effect	0.9439087
Movie & User Effect	0.8653488

3.5 Movie, User and Genre Effect Model

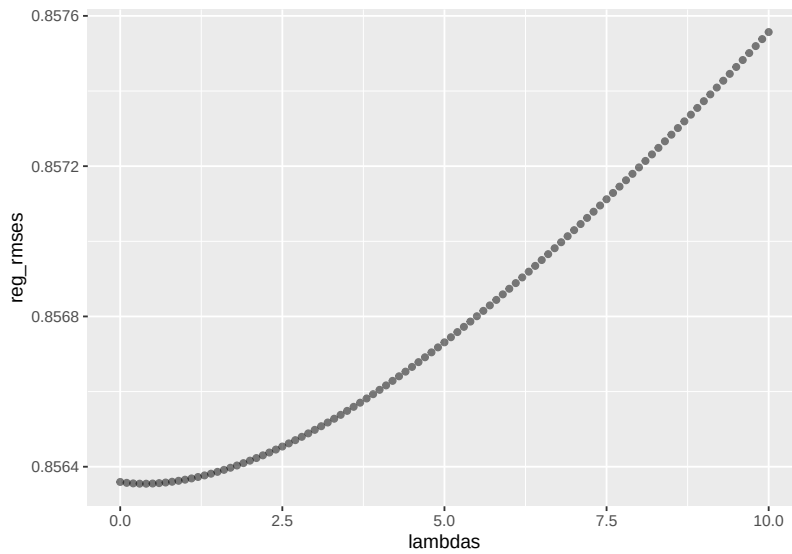
The third effect to be considered is the genre effect. Exploratory analysis (section 2.5 above) shows some genres receive generally higher ratings than others. Genre effect, b_g for a given genre combination g, is added into the model to account for the variability; and the formula to compute the genre effects:

$$Y_{u,i} = \mu + b_i + b_u + b_g + \epsilon_{ui}$$
$$\hat{b}_g = mean(\hat{y} - \hat{\mu} - \hat{b}_i - \hat{b}_u)$$

Method	RMSE
Overall rating average	1.0612018
Movie Effect	0.9439087
Movie & User Effect	0.8653488
Movie, User & Genre Effect	0.8649469

3.6 Regularized Movie, User and Genre Effect Model

The presence of extreme ratings given by a small number of users can distort the model's estimates; they can be considered as outliers. To address this, regularization is applied during algorithm training. Regularization imposes a penalty on large estimates that are formed using small sample sizes. The general idea is to constrain the overall variability to improve stability. To perform regularization, a penalty term with λ as a tuning parameter is added to the model. The penalty term regresses the estimates from small sample sizes, resulting stable estimation. To tune an optimal λ, various λs, from 0 to 10 with 0.1 intervals, are fitted into the model and used to generate the optimal value of λ that minimizes the RMSE, as shown in below:



The optimal value of λ (0.4) is then used to compute the RMSE of this model.

Method	RMSE
Overall rating average	1.0612018
Movie Effect	0.9439087
Movie & User Effect	0.8653488
Movie, User & Genre Effect	0.8649469
Regularized Movie, User & Genre Effect	0.8648472

4. Results

Below table summarizes the resulting RMSEs obtained from the models described in section 3:

Method	RMSE
Overall rating average	1.0612018
Movie Effect	0.9439087
Movie & User Effect	0.8653488
Movie, User & Genre Effect	0.8649469
Regularized Movie, User & Genre Effect	0.8648472

The final recommendation system algorithm achieved the objective RMSE value below than 0.86490.

5. Conclusion

This project developed a recommendation system algorithm using the MovieLens 10M dataset, resulting an RMSE of 0.86485, which achieves the objective of RMSE lower than 0.86490. The project first explores the dataset on relationships of movie-rating, user-rating and genre-rating. Hence movie, user and genre effects are found with evidence from the data, and the data and findings are then visualized in plots. Corresponding models are built to account for the different effects, and we can see the algorithm is progressively improved.

Finally, the algorithm is further improved by regularization of the effects. Although the algorithm here meets the project requirement, it still can be further improved. In section 2.5 we explored the genre effect, the algorithm built in this project can only consider the genre combinations, instead of individual primary genres, due to hardware limitation. Adjusting on primary genre effects would be more sophisticated and therefore lead to a more accurate result.

References

[1] https://en.wikipedia.org/wiki/Netflix_Prize/

[2] https://grouplens.org/datasets/movielens/10m/

[3] https://www.ibm.com/think/topics/loss-function

[4] https://rafalab.dfci.harvard.edu/dsbook-part-2/highdim/regularization.html#sec-netflix-loss-function

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